

HackDiwas 2.0

Crop Connect By 100xDevs

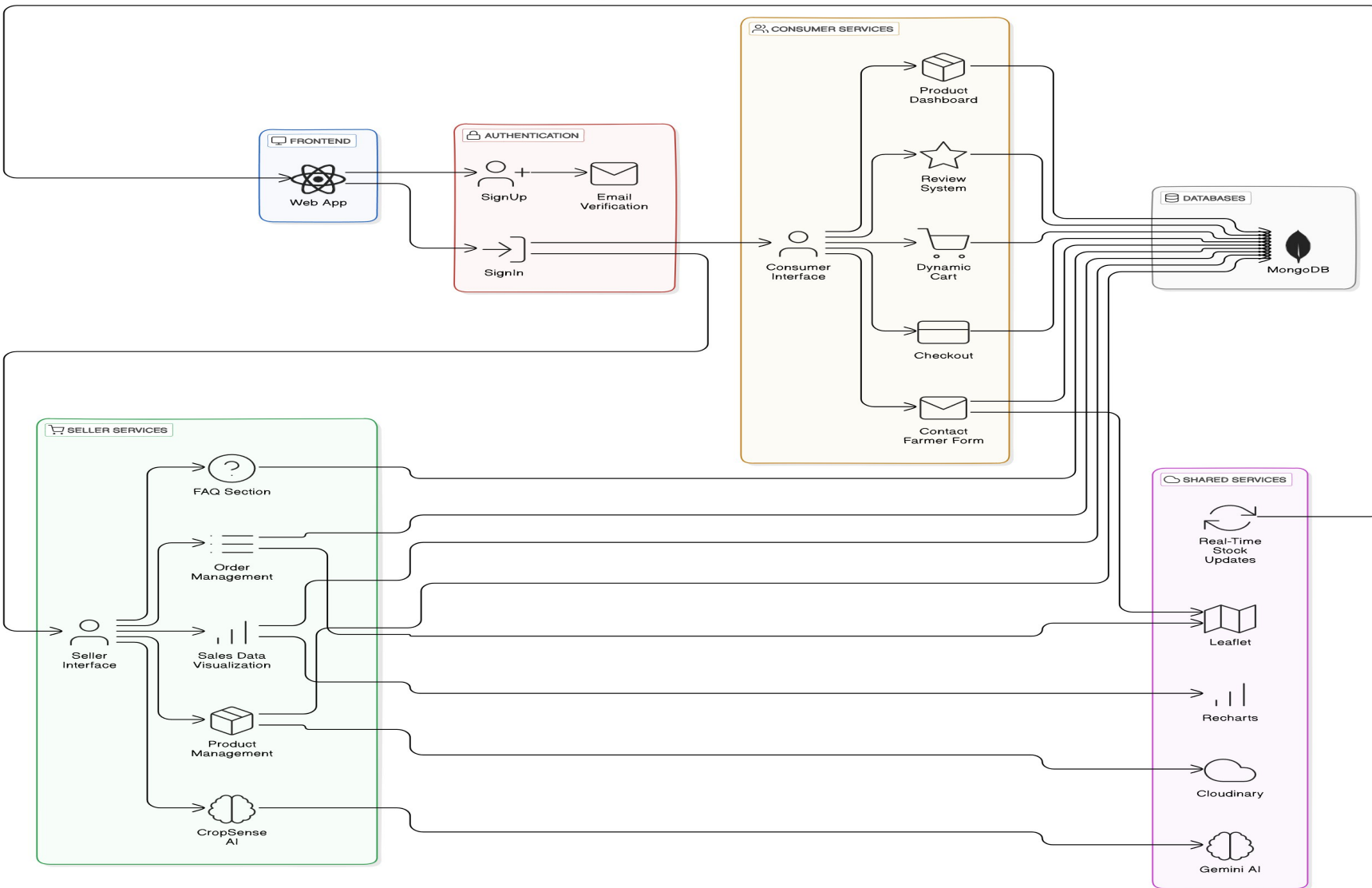
- Problem Statement ID – **HD001**
- Problem Statement Title- Sustainable Farming
- Theme-Agriculture
- PS Category- Software
- Team ID-UUHD255
- Team Name (Registered on portal) - 100XDEVS

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❖ Proposed Solution

- The Web application we are building to **directly connect the farmer with the potential buyer** to increase the income of the farmer.
- Here the farmer can list their product and directly sell them with the buyer.
- Also the farmer can sell the waste which is used in manufacturing many other items directly or indirectly.
- We also integrated the **AI(Artificial Intelligence)** to predict the best crop based on the temperature ,weather ,season and humidity to enhance the farmer income and productivity.
- Also integrated the Chat feature so that farmer and buyer can chat for the bulk order and negotiation in price.

CropConnect Architecture



Frontend:
Reactjs

Database:
MongoDB

Authentication
: JWT

AI: Gemini

Version
Control:
Git & Github

Backend:
Express



FEASIBILITY AND VIABILITY HackDiwas 2.0

1. Technical Feasibility

- **Web Platform**
 - **Front-end:** React js
 - **Back-end:** Node.js and Express js
- **AI Integration for Crop Prediction**
 - Feasible using machine learning models trained on datasets like temperature, soil type, rainfall, and seasonal trends.
 - Data sources such as IMD (Indian Meteorological Department) or other APIs (OpenWeatherMap, SoilGrids) can be used.

Chat System

- Can use tools like WebSockets to enable real-time messaging.

Database Requirements

- Cloud databases MongoDB can store user data, product listings, chat history, and weather patterns.

2. Operational Feasibility

- **Farmer Accessibility**
 - Mobile-first design ensures usability in rural areas with smartphones.
 - Can include regional language support to improve adoption.
- **Logistics Integration (optional future phase)**
 - Partnering with logistics companies or creating local pickup centers would help streamline delivery.
- **Customer Support**
 - Basic support can be managed via chatbot and regional helplines.
 - Helps bridge the tech literacy gap for some users.

3. Economic Viability

- **Revenue Generation Models**
 - **Transaction fee:** Small percentage from each transaction.
 - **Subscription model:** Premium features like analytics or visibility boosts.
 - **Ads/Promotions:** From agri-businesses, tools, seeds, fertilizers, etc.

Cost Analysis

- **Initial Development:** Web development, AI integration, and database setup (~\$10,000–\$30,000 depending on scale).
- **Ongoing Costs:** Server maintenance, customer support, updates, marketing (~\$1,000–\$3,000/month).
- **Funding Opportunities:** Government grants, agri-tech accelerators, or CSR funding from large corporations.

IMPACT AND BENEFITS

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4. Market Viability

- **Problem-Solution Fit**

- Farmers often earn less due to middlemen. This app solves that directly.
- AI helps optimize yields, directly increasing income potential.

- **Target Market**

- Primary: Small to medium farmers.
- Secondary: Urban buyers, bulk purchasers (restaurants, factories), NGOs, exporters.

- **Competitor Analysis**

- There are platforms like DeHaat, AgriBazaar, and KrishiHub—but your **direct AI crop planning + chat + waste selling feature** is a differentiator.

5. Social and Environmental Impact

- **Social Impact**

- Empowers farmers by increasing income, removing middlemen.
- Promotes digital literacy and tech adoption in rural areas.

- **Environmental Impact**

- By selling agricultural waste, you support circular economy and reduce biomass burning.
- AI helps in efficient resource utilization (water, fertilizers, etc.).

RESEARCH AND REFERENCES

- Wikipedia :- [link](#)
- Agribazar :- [link](#)
- KrishiHub :- [link](#)

Team Member

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